Mercury+ MP1 SoC Module

User Manual

Purpose

The purpose of this document is to present the characteristics of Mercury+ MP1 SoC module to the user, and to provide the user with a comprehensive guide to understanding and using the Mercury+ MP1 SoC module.

Summary

This document first gives an overview of the Mercury+ MP1 SoC module followed by a detailed description of its features and configuration options. In addition, references to other useful documents are included.

Product Information	Code	Name
Product	ME-MP1	Mercury+ MP1 SoC Module

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Document History

Version	Date	Author	Comment
02	15-Nov-2024	MGOS	Added notice about risk of damage if reprogramming the power converters. Updated block diagram. Updated list of product models. Adjusted lists of parameters for the MSS and FPGA fabric DDR4. Moved information about QSPI corruption risk to the Mercury+ MP1 SoC Module Known Issues and Changes [7]. Rewrote Section 3.7, "SPI Flash Programming from an External SPI Master". Added details about operating conditions. Improved terminology, content, and layout for better readability and consistency.
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1 Overview

1.1 General

1.1.1 Introduction

The Mercury+ MP1 SoC module combines the Microchip Polarfire® SoC (System-on-Chip) device with PCIe® Gen2 ×4, one USB 2.0 PHY (with OTG support), up to two Gigabit Ethernet PHYs, DDR4 SDRAM with Error Correction Code (ECC), eMMC flash, multi-gigabit transceivers, high-speed LVDS I/O, and is available in industrial temperature range, forming a complete and powerful embedded processing system.

The use of the Mercury+ MP1 SoC module, in contrast to building a custom SoC hardware, significantly reduces development effort and redesign risk and improves time-to-market for the embedded system.

Together with Mercury+ base boards, the Mercury+ MP1 SoC module allows the user to quickly build a system prototype and start with application development.

A Yocto layer is available for the Mercury+ MP1 SoC module. This layer allows the user to quickly build and run linux on the Mercury+ MP1 SoC module.

1.1.2 Warranty

Refer to the General Business Conditions, available on the Enclustra website [1].

Tip

The warranty of Enclustra modules is void if the FPGA's permanent lock bits are modified. This operation is irreversible. Enclustra cannot test the module in case of a product return.

NOTICE

Data loss or unusable product

PolarFire FPGAs contain permanent lock bits that can be used to secure the content of the device. Any action on any permanent lock bit is irreversible.

- Read the PolarFire Family FPGA Security User Guide [22] before performing any action on the permanent bit locks.

1.1.3 RoHS

The Mercury+ MP1 SoC module is designed and produced according to the Restriction of Hazardous Substances (RoHS) Directives 2011/65/EU (RoHS 2) and 2015/863/EU (RoHS 3).

1.1.4 Disposal and WEEE

The Mercury+ MP1 SoC module must be properly disposed of at the end of its life.

The Waste Electrical and Electronic Equipment (WEEE) Directive (2002/96/EC) is not applicable for the Mercury+ MP1 SoC module.

1.1.5 Safety Recommendations and Warnings

Mercury+ modules are not designed to be "ready for operation" for the end-user. These can only be used in combination with suitable base boards. Proper configuration of the hardware before usage is required.

Ensure that the power supply is disconnected from the board before inserting or removing the Mercury+ MP1 SoC module, connecting interfaces, or connecting jumpers.

Touching the capacitors of the DC-DC converters can lead to voltage peaks and permanent damage; over-voltage on power or signal lines can also cause permanent damage to the module.

1.1.6 Electrostatic Discharge

Electronic boards are sensitive to electrostatic discharge (ESD). Ensure that the product is handled with care and only in an ESD-protected environment.

1.1.7 Electromagnetic Compatibility

The Mercury+ MP1 SoC module is a Class A product (as defined in IEC 61000-3-2 standard) and is not intended for use in domestic environments. The product may cause electromagnetic interference, for which appropriate measures must be taken.

1.2 Features

- Microchip PolarFire SoC
  - MPFS250T/MPFS460T device
  - FCG1152 package
  - Four 64-bit RV64GC Quad Application processing cores up to 625 MHz
  - 64-bit RV64IMAC monitor processor core up to 625 MHz
  - Non-volatile FPGA fabric (28 nm process technology)
  - **MPFS250T**
    - -1 speed grade
    - 215 user I/Os
      - 208 I/Os up to 3.3 V
      - 9 MSS I/Os up to 3.3 V
    - 20 XCVRs (transceivers) @ 12.7 Gbit/s
    - 10 XCVR reference clock inputs
  - **MPFS460T**
    - standard speed grade
    - 195 user I/Os
      - 188 I/Os up to 3.3 V
      - 9 MSS I/Os
    - 16 XCVRs (transceivers) @ 10.3125 Gbit/s
    - 8 XCVR reference clock inputs
  - PCIe Gen2 ×4 support
- Up to 4 GB DDR4 SDRAM with ECC on the MSS side
- Up to 8 GB DDR4 SDRAM on the FPGA side
- 64 MB quad SPI flash
- 64 MB SPI flash
- 16 GB eMMC flash
- Up to 2 Gigabit Ethernet PHYs (SGMII interface on MSS side)
- OTG USB 2.0
- Real-time clock
- UART, SPI, I2C, SDIO/MMC

- TPM
- 5 to 13.2 V single supply

1.3 Deliverables

- Mercury+ MP1 SoC module
- Mercury+ MP1 SoC module documentation, available via download:
  - Mercury+ MP1 SoC Module User Manual (this document)
  - Mercury+ MP1 SoC Module Reference Design [2]
  - Mercury+ MP1 SoC Module IO Net Length Excel Sheet [3]
  - Mercury+ MP1 SoC Module FPGA Pinout Excel Sheet [4]
  - Mercury+ MP1 SoC Module User Schematic (PDF) [6]
  - Mercury+ MP1 SoC Module Known Issues and Changes [7]
  - Mercury+ MP1 SoC Module Footprint (Altium, Eagle, Orcad and PADS) [8]
  - Mercury+ MP1 SoC Module 3D Model (PDF) [9]
  - Mercury+ MP1 SoC Module 3D Model (STEP) [10]
  - Mercury Mars Module Pin Connection Guidelines [11]
  - Mercury Master Pinout [12]
  - Mercury Heat Sink Kits Selection and Mounting Guide [20]
- Additional resources, available online:
  - Yocto BSP Layer for Enclustra modules equipped with Microchip SoCs [17]
  - Custom Hart Software Services for Enclustra modules [18]
  - Enclustra I2C Application Note [21]

1.3.1 Reference Design

The Mercury+ MP1 SoC Module Reference Design [2] features an example configuration for the PolarFire SoC device, together with an example top level HDL file for the user logic.

A "hello world" bare-metal software application is available to provide a starting point for own user applications. Pre-compiled binaries are included in the archive, so that the user can easily check that the hardware is functional.

A custom version of the Hart Software Services [18] contains all required modifications to initialize the microprocessor subsystem and to allow booting Linux or user applications.

> **Tip**
>
> To get notified about updates to the Mercury+ MP1 SoC Module Reference Design, turn on the notifications from the Enclustra Github page [2].

1.3.2 Yocto BSP

The Enclustra Yocto BSP Layer [17] enables the user to quickly set up a Yocto project and to run Linux on the Enclustra SoC module.

The documentation describes the build process in detail and allows a user without Yocto knowledge to build and run a Linux reference design on the target hardware.

1.4 Accessories

1.4.1 Enclustra Heat Sink

For Mercury modules an Enclustra heat sink is available for purchase along with the product. Refer to Section 2.10.6 for further information on the available cooling options.

1.4.2 Mercury+ PE1 Base Board

The Mercury+ PE1 [14] is a versatile PCIe® x4 base board equipped with a multitude of I/O interfaces for use with the Mercury/Mercury+ family of FPGA and SoC modules, providing a head start for building custom FPGA and SoC based hardware systems.

It is compatible with a multitude of FMC boards from different suppliers to use in data acquisition systems, motor control, display and camera interfaces, software defined radio and more. The board is equally well suited for rapid prototyping and for building FPGA systems without designing custom hardware.

The available features depend on the Mercury module type and on the selected base board variant.

1.4.3 Mercury+ PE3 Base Board

The Mercury+ PE3 [16] is a versatile PCIe® x8 base board equipped with a multitude of I/O interfaces for use with the Mercury/Mercury+ family of FPGA and SoC modules, providing a head start for building custom FPGA and SoC based hardware systems.

This high performance base board provides a versatile set of I/O connectivity options, specialized for high-speed communication and video applications, including SFP+, QSFP+, HDMI, USB Type-C and Firefly. The board is equally well suited for rapid prototyping and for building FPGA systems without designing custom hardware.

The available features depend on the Mercury module type and on the selected base board variant.

1.4.4 Mercury+ ST1 Base Board

The Mercury+ ST1 board [15] is a compact, low-cost base board equipped with a multitude of I/O interfaces for use with the Mercury/Mercury+ family of FPGA and SoC modules.

It provides a versatile set of I/O connectivity options, specialized for video applications, including DisplayPort, HDMI, MIPI, and SFP+. The board is equally well suited for rapid prototyping and for building FPGA systems without designing custom hardware.

The available features depend on the Mercury module type and on the selected base board variant.

1.5 Microchip Tool Support

The SoC devices assembled on the Mercury+ MP1 SoC module are supported by the Microchip Libero SoC Design Suite software. A platinum license is required (support for S devices). Contact Microchip for further information.

2 Module Description

2.1 Block Diagram

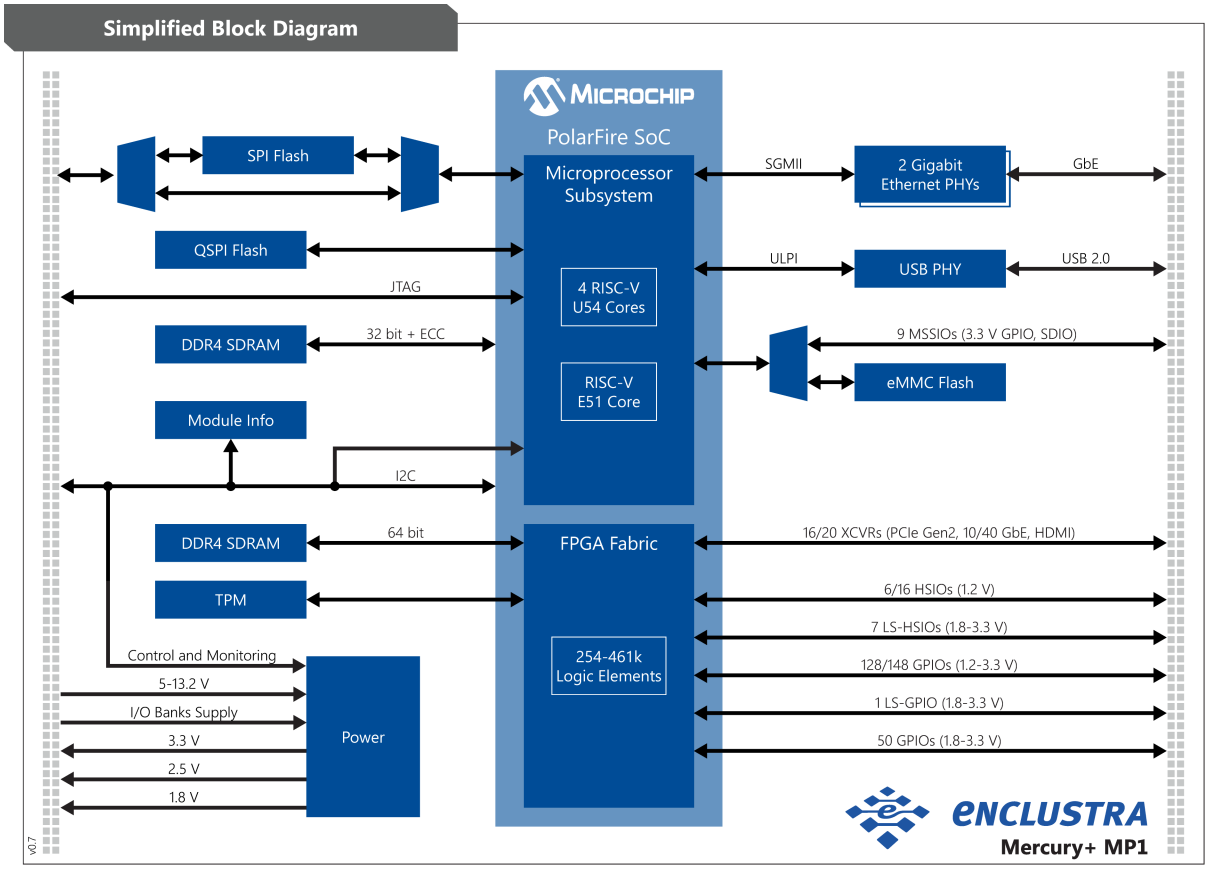

Figure 1: Hardware Block Diagram

The main component of the Mercury+ MP1 SoC module is the Microchip PolarFire SoC device. Most of its I/O pins are connected to the Mercury+ module connector, making up to 215 regular user I/Os available to the user. Further, up to 20 XCVRs are available on the module connector, enabling the implementation of several high-speed protocols such as PCIe Gen2 ×4.

Mercury+ MP1 SoC modules equipped with MPFS460 SoC devices have a different amount of XCVR transceivers and GPIOs compared to a Mercury+ MP1 SoC modules equipped with a MPFS250 SoC device. The difference between these assembly variants is described in detail in Section 2.8.2.

The FPGA is flash based and is configured almost instantly after reset is released. Alternatively the device supports loading the bitstream from SPI flash. The RISC-V processor subsystem of the Polarfire SoC allows booting from the internal flash memory. The reference design supports booting Linux from SD Card and eMMC memory. For development purposes, a JTAG interface is connected to Mercury module connector.

The available standard configurations include a 16 GB eMMC flash, a 64 MB quad SPI flash, a 64 MB SPI flash, up to 4 GB DDR4 SDRAM with ECC connected to the MSS and up to 8 GB DDR4 SDRAM connected to the FPGA.

Further, the module can be equipped with up to two Gigabit Ethernet PHYs and one USB 2.0 PHY, making

it ideal for communication applications.

A real-time clock is available on the Mercury+ MP1 SoC module.

All clock signals that are required for the assembled components are generated on the module. This includes a 125 MHz differential clock for the SGMII and MSS clocking and a 50 MHz FPGA fabric clock.

The module's internal supply voltages are generated from a single input supply of 5 to 13.2 V DC. Some of these voltages are available on the Mercury module connectors to supply circuits on the base board.

Four LEDs are connected to the SoC pins for status signaling.

2.2 Module Configuration and Product Models

Table 1 describes the available standard module configurations. The product model indicates the module type and main features. Figure 2 describes the fields within the product model. Custom configurations are available. Contact Enclustra for more information.

Product Model	SoC[1]	DDR4 ECC SDRAM (MSS)	DDR4 SDRAM (FPGA)	Ethernet PHYs
ME-MP1-250-SI-D3EN	**MPFS250**TS-FCG1152I	2 GB	N/A	2
ME-MP1-250-SI-D3EN-E1	**MPFS250**TS-FCG1152I	2 GB	N/A	1
ME-MP1-460-1SI-D4E[2]	**MPFS460**TS-1FCG1152I	4 GB	8 GB	2

Table 1: Standard Module Configurations

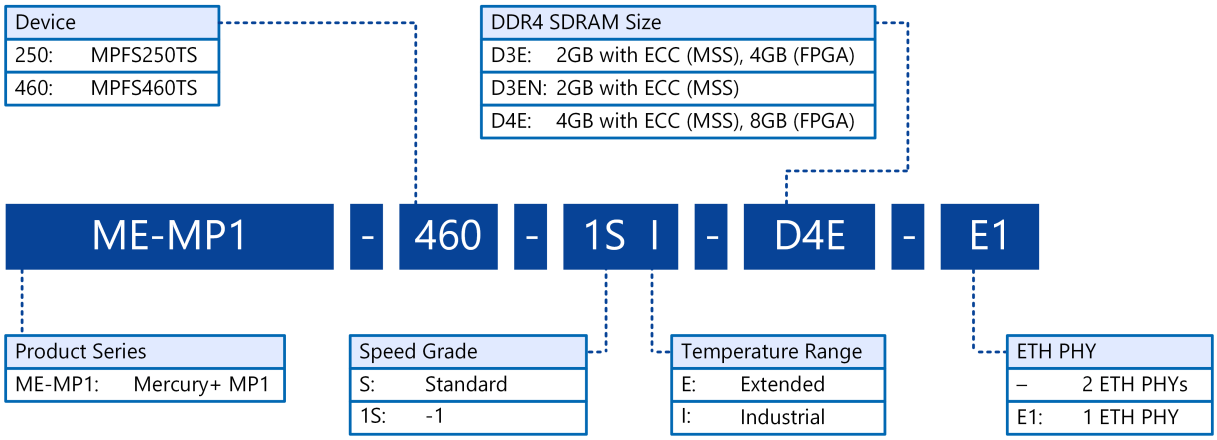

Figure 2: Product Model Fields

For the first revision modules or early access modules, the product model may not respect entirely this naming convention. Contact Enclustra for more information.

[1]The "device" is shown in bold.
[2]Under development

2.3 EN-Numbers and Product Models

Every product is uniquely labeled, showing the EN-number and serial number. An example is presented in Figure 3.

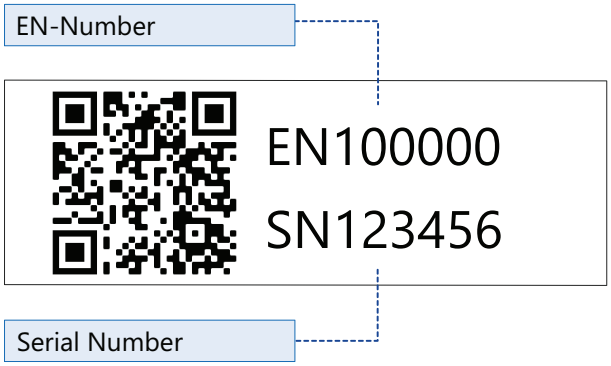

Figure 3: Module Label

The correspondence between EN-number and product model for each revision is shown in Table 2.

The known issues of the product and the changes between the revisions are described in the Mercury+ MP1 SoC Module Known Issues and Changes [7].

EN-Number	Product Model	Revision Number
EN105249	ME-MP1-250-SI-D3EN-E1	R1.0
EN104234	ME-MP1-250-SI-D3EN	R1.0

Table 2: EN-Numbers and Product Models

2.4 Top and Bottom Views

Depending on the hardware revision and configuration, the module may look slightly different than shown in this document.

2.4.1 Top View

Figure 4: Module Top View

2.4.2 Bottom View

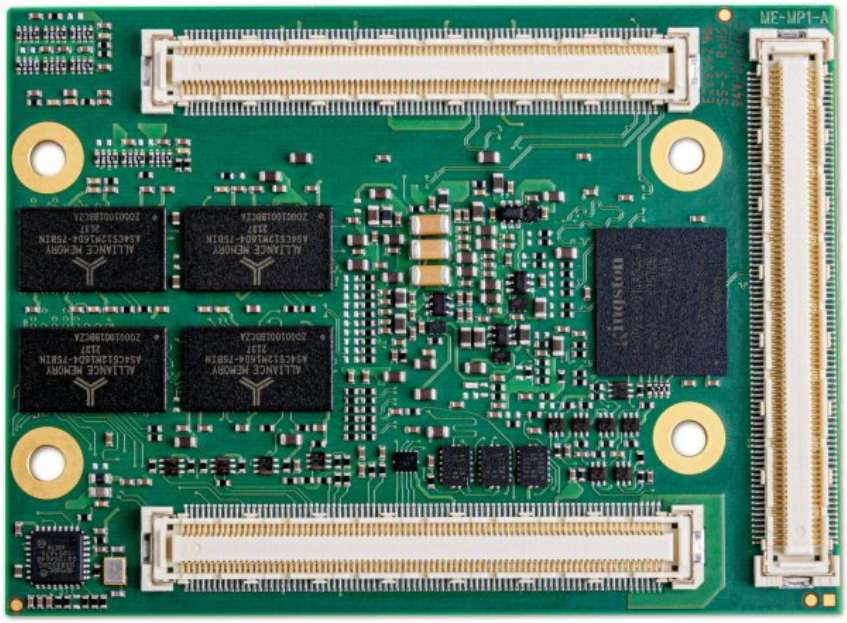

Figure 5: Module Bottom View

2.5 Top and Bottom Assembly Drawings

Depending on the hardware revision and configuration, the module may look slightly different than shown in this document.

2.5.1 Top Assembly Drawing

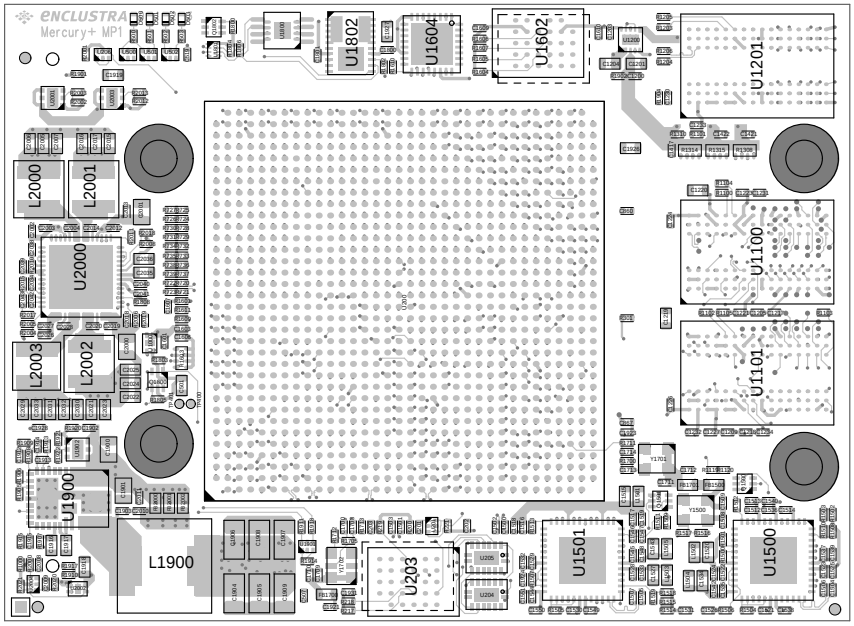

Figure 6: Module Top Assembly Drawing

2.5.2 Bottom Assembly Drawing

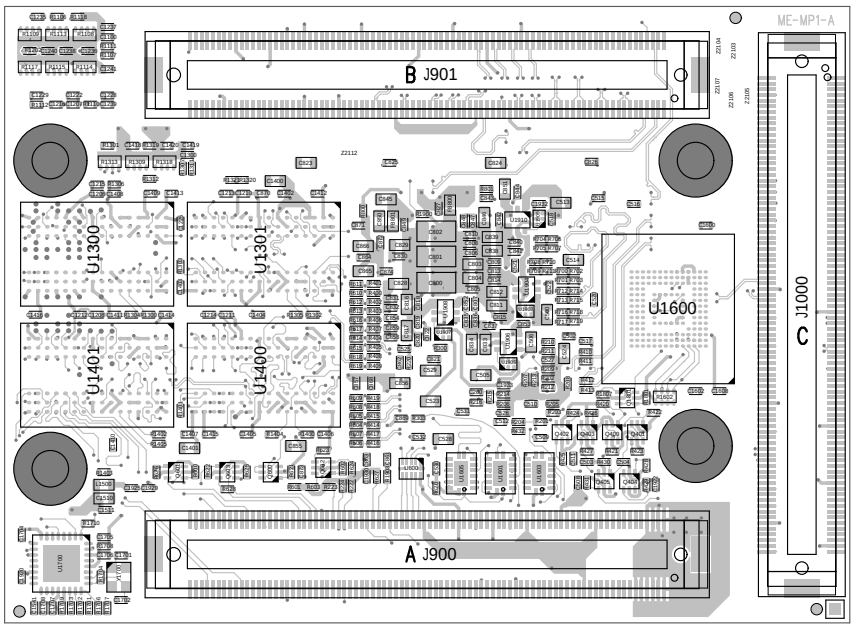

Figure 7: Module Bottom Assembly Drawing

2.6 Module Footprint and Mechanical Data

Figure 8 shows the dimensions of the module footprint on the base board.

Enclustra offers Mercury and Mercury+ modules of various geometries having widths of 56, 64, 65, 72 or 74 mm and having different topologies for the mounting holes. If different module types shall be fixed on the base board by screws, additional mounting holes may be required to accommodate different modules. The footprints of the module connectors for the base board design are available for different PCB design tools (Altium, PADS, Eagle, Orcad) [8] and include the required information on the module sizes and holes.

The maximum component height under the module is dependent on the connector type. Refer to Section 2.7 for detailed connector information.

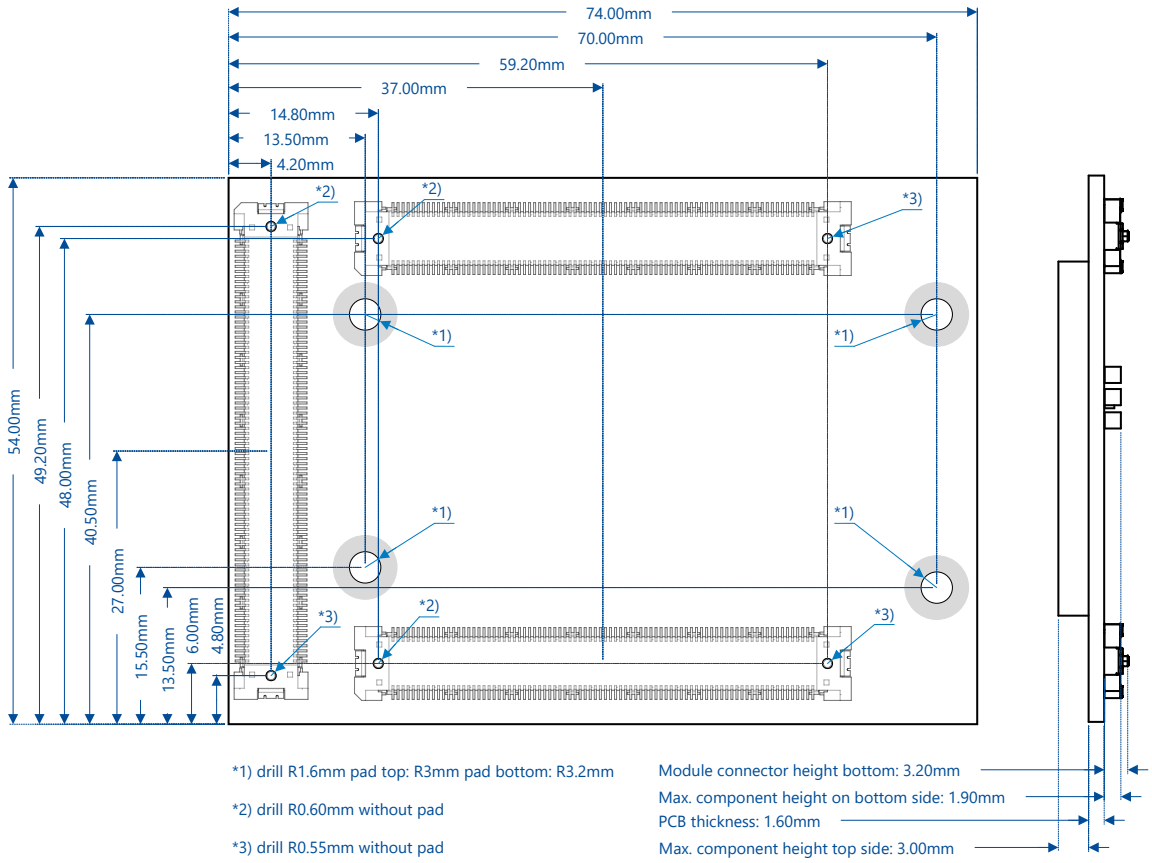

Figure 8: Module Footprint and Dimensions - Bottom View and Side View

Tip

A small golden square on the bottom left corner of Enclustra modules is provided as landmark. The same landmark is provided on the Enclustra base boards to help orient the module in the right direction when connecting it.

NOTICE

Damage to the connectors

The connectors of the module and the base board can be damaged if the connectors are not properly aligned during installation.

- Align the connectors carefully before applying force on the module.
- Do not use excessive force to latch the module into the connectors on the base board.

NOTICE

Damage to the device when applying power

The Mercury module could physically be mounted the wrong way around on the base board. The module and the base board can be damaged if the device is powered on while it is connected the wrong way around.

- Ensure that the mounting holes on the base board are aligned with the mounting holes of the module.

Table 3 describes the mechanical characteristics of the Mercury+ MP1 SoC module. A 3D model (PDF) and a STEP 3D model are available [9], [10].

Parameter	Value
Size	74 × 54 mm
Component height top	3.00 mm
Component height bottom	1.9 mm
Weight	36 g

Table 3: Mechanical Data

2.7 Module Connector

Three Hirose FX10 168-pin 0.5 mm pitch headers with a total of 504 pins have to be integrated on the base board [13]. Up to four M3 screws may be used to mechanically fasten the module to the base board. Do not use excessive force to tighten the screws, as this could damage the module.

The pinout of the module connector is found in the Mercury Master Pinout [12]. The connector is available in different packaging options and different stacking heights. Some examples are presented in Table 4. Refer to the connector datasheet for more information.

Component	Part Number	Description
Mercury module connector	FX10A-168S-SV	Hirose FX10, 168-pin, 0.5 mm pitch
Base board connector	FX10A-168P-SV(71)	Hirose FX10, 168-pin, 0.5 mm pitch, 4 mm stacking height
Base board connector	FX10A-168P-SV1(71)	Hirose FX10, 168-pin, 0.5 mm pitch, 5 mm stacking height

Table 4: Module Connector Characteristics

Figure 9 indicates the pin numbering for the Mercury module connectors from the top view of the base board. The connector pins are numbered as follows:

- Connector A: from J900-1 to J900-168
- Connector B: from J901-1 to J901-168
- Connector C: from J1000-1 to J1000-168

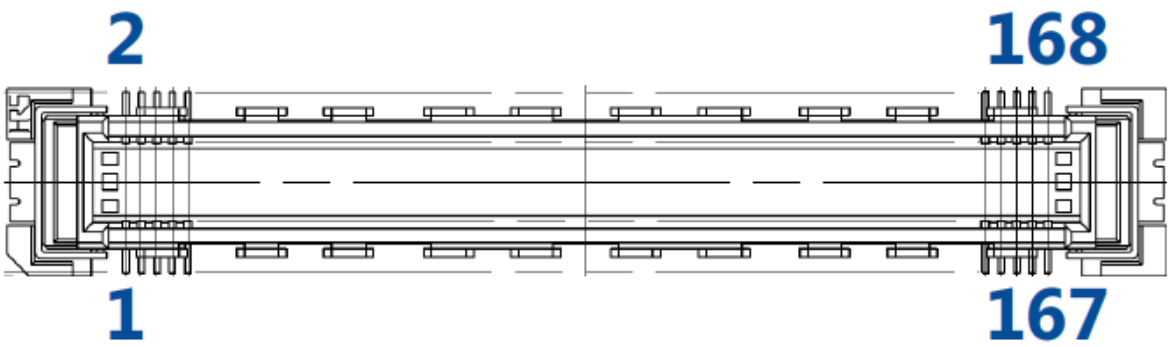

Figure 9: Pin Numbering for the Module Connector

2.8 User I/O

2.8.1 Pinout

Information on the Mercury+ MP1 SoC module pins can be found in the Mercury Master Pinout [12], and in the additional document Mercury Mars Module Pin Connection Guidelines [11].

> **Tip**
>
> The pin types on the schematic of the module connector and in the master pinout document are for reference only. On the Mercury+ MP1 SoC module, the connected pins might not have the targeted functions (such as primary clocks, differential pins, transceiver signals, etc).

> **Tip**
>
> The availability of certain pins might depend on the product model.

The naming convention for the user I/Os located on GPIO banks is:
GPIO_B<BANK>_T<GROUP>_L<PAIR_NUMBER>_<SPECIAL_FUNCTION>_<POLARITY>.

For example, GPIO_B1_T3_L4_CLKIN_P is pair 4 of group 3 and is connected to FPGA GPIO bank 1. It has positive polarity if used as a differential pair with the special function of a preferred clock input.

IO_B1_C8_LS and BOOT_MODE_LS are an exception and do not follow this naming convention. It is connected through a level shifter to GPIO Bank 1. It is also separately mentioned in chapter 2.8.2.

The naming convention for the user I/Os located on HSIO banks is:
IO_B<BANK>_<PIN_NAME>_<SPECIAL_FUNCTION>.

For example, IO_B8_AN4_UARTRX is connected to FPGA HSIO bank 8, is connected to the SoC at pin AN4 and can be used for an UART interface.

FPGA_DONE_LS is an exception and does not follow this naming convention. It is connected through a level shifter to HSIO Bank 0. It is also separately mentioned in chapter 2.8.2.

Further special functions are listed in the Polarfire SoC FPGA Packaging and Pin Descriptions [24].

Table 5 lists information related to the total number of I/Os available in each I/O bank and possible limitations.

Signal Name	Number of Signals	Number of Pairs	Differential	Single-ended	Bank	Bank Type
GPIO_B1_<...>	56	28	In/Out	In/Out	1	GPIO
IO_B1_C8_LS	1	0	In/Out	In/Out	1	GPIO
BOOT_MODE	1	0	In/Out	In/Out	1	GPIO
GPIO_B7_<...>	50	25	In/Out	In/Out	7	GPIO
GPIO_B9_<...>	92 (MPFS250) 72 (MPFS460)	46 (MPFS250) 36 (MPFS460)	In/Out	In/Out	9	GPIO
IO_B0_<...>	5	0	In/Out	In/Out	0	HSIO
FPGA_DONE_LS	1	0	In/Out	In/Out	0	HSIO
IO_B8_<...>	2	0	In/Out	In/Out	8	HSIO

Table 5: User I/Os

The multi-gigabit transceiver (XCVR) are described in Section 2.9.

2.8.2 I/O Pin Exceptions

The I/O pin exceptions are pins with special functions or restrictions (for example, when used in combination with certain Mercury boards they may have a specific role).

UART Interface

Table 6 lists the signals on the Mercury+ MP1 SoC module related to the UART interface, which is connected via a level shifter.

Signal Name	Connector Pin	SoC Bank	SoC Pin	Connector Voltage Level	SoC Voltage Level
IO_B8_AN4_UARTRX	A-105	8	AN4	VCC_CFG	VCC_1V2
IO_B8_AP4_UARTTX	A-107	8	AP4	VCC_CFG	VCC_1V2

Table 6: I/O Pin Exceptions - UART interface

Both signals have a level of VCC_CFG on the connector and are connected to the FPGA via a level shifter to 1.2 V.

I/O Pins with Level Shifter

Some signals on the Mercury+ MP1 SoC module are routed from the FPGA banks to the module connector via level shifters. These are presented in Table 7.

Signal Name	Connector Pin	SoC Bank	SoC Pin	Connector Voltage Level	SoC Voltage Level
IO_B0_AN26_LS	A-88	0	AN26	VCC_CFG	VCC_1V2
IO_B0_AN27_LS	A-90	0	AN27	VCC_CFG	VCC_1V2
IO_B0_AL25_LS	A-92	0	AL25	VCC_CFG	VCC_1V2
IO_B0_AM24_LS	A-94	0	AM24	VCC_CFG	VCC_1V2
IO_B1_C8_LS	A-110	1	C8	VCC_CFG	VCC_IO_B1
BOOT_MODE_LS	A-112	1	C7	VCC_CFG	VCC_IO_B1
FPGA_DONE_LS	A-130	0	AL24	VCC_CFG	VCC_1V2
IO_B0_AE25_PERST#_LS	A-104	0	AE25	VCC_CFG	VCC_1V2

Table 7: I/O Pin Exceptions - Level Shifters

The level shifters used for signals IO_B0_AN26_LS, IO_B0_AN27_LS, IO_B0_AL25_LS and IO_B0_AM24_LS are NXP NTB0104 and the maximum achievable data rate on these pins is 30 Mbit/sec. The signal level of IO_B1_C8_LS is shifted by a EM6K34 MOSFETs.

Assembly Options

The number of user I/Os and XCVRs available on the module connector depends on the SoC device assembled on the Mercury+ MP1 SoC module.

On the Mercury+ MP1 SoC modules equipped with the bigger SoC device (MPFS460), there are four XCVR transceiver lines with two differential clock inputs more available than on the Mercury+ MP1 SoC modules equipped with the smaller SoC device (MPFS250). Mercury+ MP1 SoC modules equipped with the smaller SoC device (MPFS250) have 20 GPIOs more instead.

The diagram shown in Figure 10 shows how the shared signals on connector C are connected via resistor assembly options to the FPGA for the two different SoC devices. More details are available in the Mercury+ MP1 SoC Module FPGA Pinout Assembly Variants Excel Sheet [5].

Design support files such as the Mercury Master Pinout [12], Mercury+ MP1 SoC Module User Schematic [6], and Mercury+ MP1 SoC Module FPGA Pinout Assembly Variants Excel Sheet [5] offer additional in-

formation on assembly options. In the Mercury+ MP1 SoC Module User Schematic [6], this information is depicted in the "Assembly Variants" section at the end of the PDF file.

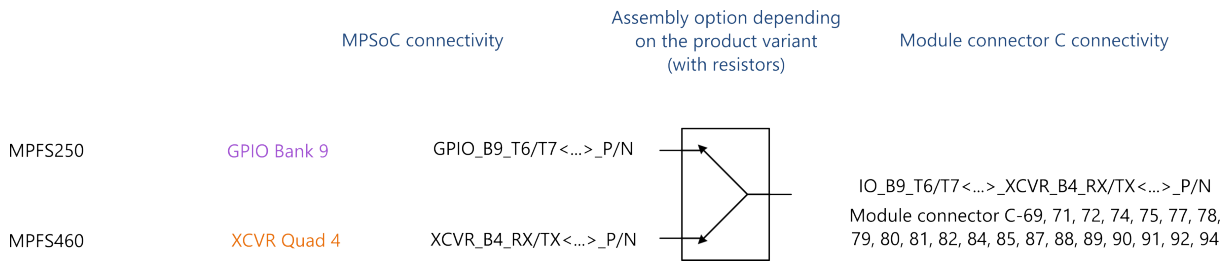

Figure 10: Signal connection for different SoC types.

Additionally, there are 10 GPIOs (GPIO_B1_T3_L[1-5]_P/N) on the module connector that are connected to GPIO Bank 1. If the Mercury+ MP1 SoC module is equipped with a MPFS460 SoC device, these signals are additionally routed to HSIO Bank 0 via a 49.9 Ω resistor.

2.8.3 Differential I/Os

When using differential pairs, a differential impedance of 100 Ω must be matched on the base board, and the two nets of a differential pair must have the same length.

The information regarding the length of the signal lines from the SoC device to the module connector is available in Mercury+ MP1 SoC Module IO Net Length Excel Sheet [3]. This enables the user to match the total length of the differential pairs on the base board if required by the application.

> **Tip**
>
> The trace length of various signals may change between revisions of the Mercury+ MP1 SoC module. Use the information provided in the Mercury+ MP1 SoC Module IO Net Length Excel Sheet [3] to check which signals are affected. The differential signals will also be routed differentially in subsequent product revisions.

While GPIO pins have an internal 100 Ω differential termination resistor that can be enabled in the Libero SoC software, HSIO pins need external termination.

> **Tip**
>
> Before producing your own base board, validate the Mercury+ MP1 SoC module pinout with Libero SoC Design Suite to make sure that all pins are used according to the correct direction.

2.8.4 I/O Banks

Table 8 describes the main attributes of the FPGA fabric and Microprocessor Subsystem (MSS) I/O banks, and indicates which peripherals are connected to each I/O bank. All I/O pins within a particular I/O bank must use the same I/O supply voltage (VCC_IO).

Bank	Bank Type	Connectivity	Supply Voltage
XCVR Banks			
0	XCVR	Module connector	1.05 V
1	XCVR	Module connector	1.05 V

Continued on next page...

Bank	Bank Type	Connectivity	Supply Voltage
2	XCVR	Module connector	1.05 V
3	XCVR	Module connector	1.05 V
4	XCVR	Module connector	1.05 V
FPGA Banks			
0	HSIO	FPGA DDR4 SDRAM, module connector	1.2 V
8	HSIO	FPGA DDR4 SDRAM, clock oscillator, module connector	1.2 V
1	GPIO	Module connector, I2C, Ethernet PHY GPIO	VCC_IO_B1, user selectable
7	GPIO	Module connector, TPM, Power Synchronization	VCC_IO_B7, user selectable
9	GPIO	Module connector, LEDs	VCC_IO_B9, user selectable
MSS Banks			
2	MSS	I2C, QSPI, LEDs, MDIO, USB	VCC_3V3_SEQ
4	MSS	SDIO, eMMC	VCC_CFG
5	MSS	SGMII (Ethernet)	VCC_3V3_SEQ
6	MSS DDR	MSS DDR4 SDRAM	1.2 V
3	Configuration	JTAG, SPI Flash, External SPI master	VCC_CFG

Table 8: I/O Banks

2.8.5 VCC_IO Usage

The VCC_IO voltages for the I/O banks located on the module connector are configurable by applying the required voltage to the VCC_IO_B[x], respectively VCC_CFG_[x] pins. All VCC_IO_B[x] or VCC_CFG_[x] pins of the same bank must be connected to the same voltage.

For compatibility with other Enclustra Mercury modules, it is recommended to use a single I/O voltage per module connector.

Signal Name	SoC Pins	Supported Voltages	Connector A Pins	Connector B Pins	Connector C Pins
VCC_CFG	VDDI3[3], VDDI4	1.8 V - 3.3 V ±5%	74, 77	-	-
VCC_IO_B7	VDDI7	1.8 V - 3.3 V ±5%	38, 41	-	-
VCC_IO_B9	VDDI9	1.2 V - 3.3 V ±5%	-	64, 67, 88, 95, 140, 143	-
VCC_IO_B1	VDDI1	1.2 V - 3.3 V ±5%	-	-	76, 116, 158

Table 9: VCC_IO Pins

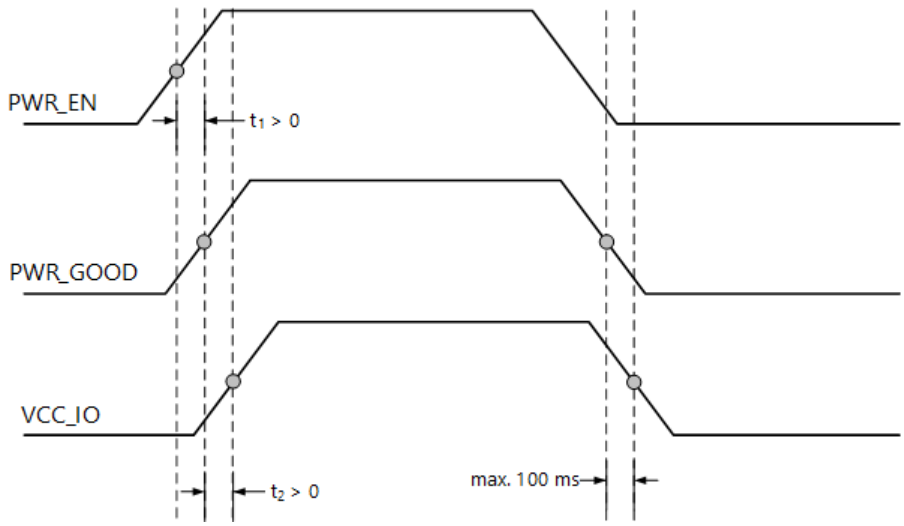

Figure 11: Power-Up Sequence - VCC_IO in Relation with PWR_GOOD and PWR_EN Signals

NOTICE

Damage to the device due to unsuitable voltage

Unsuitable voltages may damage the SoC device as well as other devices on the Mercury+ MP1 SoC module.

- Only use VCC_IO voltages compliant with the assembled SoC device.

NOTICE

Damage to the device due to floating VCC_IO pins

Floating VCC_IO pins reduce ESD protection.

- Do not leave any VCC_IO pin floating.

[3] Microchip updated the power-up requirements for VDDI3 [29]. More details about this issue are available in the Mercury+ MP1 SoC Module Known Issues and Changes [7]

⚠ CAUTION

Injury due to uncontrolled device

If an I/O is connected to an external device, violating the power sequence or leaving the corresponding VCC_IO pin floating will leave the pin in an undefined state. This can lead to any behavior of the devices attached to this pin and, potentially, to damage or injuries.

- Follow the power sequence diagram shown in Figure 11. This ensures that the I/Os are tri-stated at power-on and power-off.
- Do not leave VCC_IO pins floating.

2.8.6 Signal Terminations

Differential I/Os

There are no external differential termination resistors on the Mercury+ MP1 SoC module for differential I/Os. These pairs may be terminated by external termination resistors on the base board (close to the module pins).

For available on-die differential termination resistors, refer to the PolarFire FPGA and SoC FPGA User I/O User Guide [27].

Single-Ended I/Os

There are external 49.9 Ω series termination resistors on some signals These resistors are required for the implementaion of MIPI interfaces. For details, refer to the PolarFire SoC FPGA Board Design Guidelines [26] and the Mercury+ MP1 SoC Module User Schematic [6].

For available on-die differential termination resistors, refer to the PolarFire FPGA and SoC FPGA User I/O User Guide [27].

2.8.7 MSS I/O Pins

Table 10 gives an overview over the MSS I/O pin connections on the Mercury+ MP1 SoC module.

Pins	Function(s)	Connection(s)
[0:6]	eMMC flash SD card	eMMC SD card
[7:8]	eMMC flash User functionality	eMMC External user functionality via multiplexer
[9]	eMMC flash SD card	eMMC SD card
[10:11]	eMMC flash User functionality	eMMC External user functionality via multiplexer
[12]	MSS reset (MSS_RST#)	SPI Flash, Ethernet PHY(s), USB PHY, QSPI and eMMC flash reset
[13]	I2C interrupt	Module connector via level shifter
[14:25]	USB	USB 2.0 PHY
[26:27]	I2C	Module connector via level shifter
[28:29]	Ethernet MDIO	Gigabit Ethernet PHY 0 and 1
[30:35]	QSPI flash	QSPI flash
[36:37]	LED0#, LED1#	On-board LEDs via buffer

Table 10: MSS I/O Pins Connections Overview

2.9 Transceiver (XCVR)

The transceivers on the Mercury+ MP1 SoC module are referred to as XCVR and are all connected to the FPGA of the SoC.

There are 16 XCVR transceivers available, which are organized in four quads.

Eight XCVR pairs and two corresponding clocks are routed to module connectors B and C, respectively. This results in a total amount of 16 XCVR pairs (64 XCVR signals) and 8 clocks (16 XCVR clock signals) available on the module connectors.

Additionally, modules equipped with a MPFS460T device have a fifth quad available on connector C. Other devices have GPIOs on these connector pins. Table 11 describes the connections.

Signal Name	Signal Description	Pairs	Quad	Comment
XCVR_B0_RX<...>	XCVR receivers	4	0	-
XCVR_B0_TX<...>	XCVR transmitters	4		
XCVR_B0_REFCLK[A,B]<...>	XCVR reference input clocks	2		
XCVR_B1_RX<...>	XCVR receivers	4	1	-
XCVR_B1_TX<...>	XCVR transmitters	4		
XCVR_B1_REFCLK[A,B]<...>	XCVR reference input clocks	2		
XCVR_B2_RX<...>	XCVR receivers	4	2	-
XCVR_B2_TX<...>	XCVR transmitters	4		
XCVR_B2_REFCLK[A,B]<...>	XCVR reference input clocks	2		
XCVR_B3_RX<...>	XCVR receivers	4	3	-
XCVR_B3_TX<...>	XCVR transmitters	4		
XCVR_B3_REFCLK[A,C]<...>	XCVR reference input clocks	2		
XCVR_B4_RX<...>	XCVR receivers	4	4	Only on MPFS460 devices
XCVR_B4_TX<...>	XCVR transmitters	4		
XCVR_B4_REFCLK[A,B]<...>	XCVR reference input clocks	2		

Table 11: XCVR Pairs

The naming convention for the XCVR I/Os is:
XCVR_B<QUAD_NUMBER>_<FUNCTION><PAIR_NUMBER_LETTER>_<POLARITY>.

For example, XCVR_B1_RX0_P is the first receiver signal of quad 1 with positive polarity.

The XCVR signals on a standard speed grade SoC device support data rates of up to 10.3125 Gbit/s whereas devices with speed grade -1 support data rates of up to 12.7 Gbit/s.

The SoC devices assembled on the Mercury+ MP1 SoC module can support up to one integrated PCIe Gen3 ×4 interfaces on the FPGA fabric, implemented using XCVR transceivers. This interface is only available within quad 0. For more information, refer to the PolarFire FPGA and PolarFire SoC FPGA PCI Express User Guide [23].

Tip

For optimal performance of high-speed interfaces, for example, PCIe, use redrivers on the base board.

Tip

The maximum data rate on the transceivers on the Mercury+ MP1 SoC module depends on the routing path for these signals. When using transceivers at high performance rates, ensure adequate signal integrity over the full signal path.

NOTICE

Damage to the transceivers

No AC coupling capacitors are placed on the Mercury+ MP1 SoC module on the transceivers.

- If required by your application, ensure that capacitors are mounted on the base board, close to the module pins.

2.10 Power

2.10.1 Power Generation Overview

The Mercury+ MP1 SoC module uses a 5 to 13.2 V DC power input for generating the on-board supply voltages. The power output pins of the module are accessible on the module connector.

Table 12 describes the power supplies generated on the module.

Generated Supply Name	Voltage Value	Rated Current	Source Supply Name	Shut down via PWR_EN	Influences PWR_GOOD
VCC_VDD	1.05/1 V (Core power supply)	20 A	VCC_MOD	Yes	Yes
VCC_1V2	1.2 V	4 A	VCC_MOD	Yes	Yes
VCC_1V8	1.8 V	2 A	VCC_MOD	Yes	Yes
VCC_2V5	2.5 V	2 A	VCC_MOD	Yes	Yes
VCC_3V3	3.3 V	4 A	VCC_MOD	No	Yes
VREF_2V2	2.2 V	0.5 A	VCC_2V5	Yes	Yes
VCC_3V3_PI	3.3 V	0.05 A	VCC_MOD	No	No
VCC_5V0_IRPS	5.0 V	0.05 A	VCC_MOD	No	No
VCC_5V0_IR	5.0 V	0.05 A	VCC_MOD	No	No

Table 12: Generated Power Supplies

In the standard configuration the core power supply is 1.05 V. For custom configurations this voltage can be changed to 1 V.

Refer to the Mercury Mars Module Pin Connection Guidelines [11] for general rules on the power pins.

NOTICE

Damage to the module or the base board

Reprogramming the power converters can cause irreversible damage to the module or the base board. Any attempt to reprogram the power converters causes the warranty to be rendered void.

- Do not reprogram the power converters.

Power Converter Synchronization

All switching DC/DC converters used on the Mercury+ MP1 SoC module support synchronization of the switching frequency with an external clock. The module includes a power synchronization circuit, which may drive a clock generated by the SoC device to all the DC/DC converters.

By default, the free-running switching frequency of the DC/DC converters is set. To set the switching frequency to a desired value, a clock must be generated on SoC pin M6 of bank 7. Table 13 presents the control signal for the power converter synchronization.

Signal Name	SoC Pin	Description
PWR_SYNC	M6	Clock signal used for DC/DC converters synchronization • Connect to GND or leave floating if power synchronization is not needed. • Clock signal: The DC/DC converters are synchronized with this external clock signal.

Table 13: Power Converter Synchronization

The synchronization frequency valid for all switching DC/DC converters on-board and compensation networks must lie in the range of 600 kHz ± 3% for the whole operating temperature range of -40 to 125°C. For a temperature range of 0 to 85°C this frequency range increases to 600 kHz ± 6.25%.

2.10.2 Power Enable/Power Good

The Mercury+ MP1 SoC module provides a power enable input on the module connector. This input may be used to shut down the DC/DC converters and LDOs for 1.05/1 V, 1.2 V, 1.8 V, 2.2 V and 2.5 V. The list of regulators that can be disabled via PWR_EN signal is provided in Section 2.10.1.

The PWR_EN input and PWR_GOOD signal are pulled to VCC_3V3 on the Mercury+ MP1 SoC module with a 4.7 kΩ resistor.

PWR_GOOD is an open collector signal and must not be used to drive a load directly. This signal is pulled to GND if the on-board regulators fail or if the module is disabled via PWR_EN. The list of regulators that influence the state of PWR_GOOD signal is provided in Section 2.10.1.

Pin Name	Connector Pin	Comment
PWR_EN	A-10	Floating/3.3 V: Module power enabled Driven low: Module power disabled
PWR_GOOD	A-12	0 V: Module supply not ok 3.3 V: Module supply ok

Table 14: Module Power Status and Control Pins

NOTICE

Damage to the device due to unsuitable voltage

Appling unsuitable voltage to the PWR_EN pins can damage the Mercury+ MP1 SoC module.

- Do not apply any other voltages to the PWR_EN pins than 3.3 V or GND.
- PWR_EN pins can be left unconnected.

2.10.3 Voltage Supply Inputs

Table 15 describes the power supply inputs on the Mercury+ MP1 SoC module. The VCC voltages used as supplies for the I/O banks are described in Section 2.8.5.

Supply Name	Connector Pins	Voltage	Description
VCC_MOD	A-1, 2, 3, 4, 5, 6, 7, 8, 9, 11	5 - 13.2 V	Supply for on-module generated power supplies. The 2.2 V supply is generated from the 2.5 V supply. The input current is rated at 3 A (0.3 A per connector pin).
VCC_BAT	A-168	1.8 - 5.5 V	Battery voltage for battery-backed RAM and battery-backed RTC

Table 15: Voltage Supply Inputs

2.10.4 Voltage Supply Outputs

Table 16 presents the supply voltages generated on the Mercury+ MP1 SoC module, that are available on the module connector.

Supply Name	Connector Pins	Voltage	Maximum Current[4]	Comment
VCC_3V3	A-26, 29, 50, 86 B-55, 79, 115, 127, 152, 155 C-96, 103, 136, 143	3.3 V ±5%	1.5 A (and max 0.3 A per pin)	Always active
VCC_2V5	A-53, 62, 65, 89	2.5 V ±5%	0.5 A	Controlled by PWR_EN
VCC_1V8	B-52, 76, 108, 128 C-83, 123, 165	1.8 V ±5%	1 A	Controlled by PWR_EN

Table 16: Voltage Supply Outputs

NOTICE

Damage to the device due to unsuitable usage of the output pins

- Do not connect any power supply to the voltage supply outputs.
- Do not short circuit any of the voltage supply outputs to GND.

2.10.5 Power Consumption

The power consumption of any SoC device strongly depends on the application (on the configured bitstream and I/O activity).

To estimate the power consumption of your design, use the Microchip PolarFire and PolarFireSoc Power Estimator and Power Analyzer available on the Microchip website.

2.10.6 Heat Dissipation

High performance devices like the Microchip PolarFire SoC need cooling in most applications; always make sure the SoC is adequately cooled.

For Mercury modules an Enclustra heat sink kit is available for purchase along with the product. It represents an optimal solution to cool the Mercury+ MP1 SoC module - the heat sink body is low profile and usually covers the whole module surface. The kit comes with a gap pad for the SoC device, a fan and required mounting material to attach the heat sink to the module PCB and baseboard PCB. With additional user configured gap pads, it is possible to cool other components on the board as well.

Alternatively, if the Enclustra heat sink does not match the application requirements, a third-party heat sink body (ATS) and an additional gap pad (t-Global) may be used. The Enclustra heat sink kit already contains all necessary items for cooling the module (heat sink body, gap pad, fan, mounting material).

Table 17 lists the heat sink and thermal pad part numbers that are compatible with the Mercury+ MP1 SoC module. Details on the Mercury heat sink kit can be found in the Mercury Heat Sink Kits Selection and Mounting Guide [20].

[4]The maximum available output current depends on your design. See Sections 2.10.1 and 2.10.5 for details.

Product Name	Package Name	Enclustra Heat Sink	ATS Heat Sink	t-Global Thermal Pad
Mercury+ MP1	FCG1152 [24]	ACC-HS4-Set	ATS-52350G-C1-R0	TG-A6200-28-28-1

Table 17: Heat Sink Characteristics

> **Tip**
>
> The adhesive heat sink part is recommended only for prototyping purposes. When the module is used in environments subject to vibrations, additional mechanical fixation is recommended.

NOTICE

Damage to the device due to overheating

Depending on the user application, the Mercury+ MP1 SoC module may consume more power than can be dissipated without additional cooling measures.

- Ensure that the SoC is always adequately cooled by installing a heat sink and/or providing air flow.

2.10.7 Voltage Monitoring

Several pins on the module connector on the Mercury+ MP1 SoC module are marked as VMON. These are voltage monitoring outputs that are used in the production test for measuring some of the on-board voltages.

It is not allowed to draw power from the voltage monitoring outputs.

Table 18 presents the VMON pins on the Mercury+ MP1 SoC module.

> **Tip**
>
> The voltage monitoring outputs are for Enclustra-use only. Pinout changes may be applied between revisions.

Signal Name	Connector Pin	Connection	Description
VMON_1V2	A-102	VCC_1V2	1.2 V on-board voltage
VMON_VTT	B-8	VCC_VTT	DDR termination voltage
VMON_VDD	B-168	VCC_VDD	Core voltage
VMON_2V5	B-167	VCC_2V5	2.5 V on-board voltage
VMON_1V8	C-8	VCC_1V8	1.8 V on-board voltage

Table 18: Voltage Monitoring Outputs

2.11 Clock Generation

A 50 MHz oscillator is used for the Mercury+ MP1 SoC module clock generation; the 50 MHz clock is fed to the HSIO bank 8 (pin AM5). A 125 MHz LVDS oscillator is connected to MSS.

A 24 MHz clock and a 25 MHz clock are used for the USB PHY and Ethernet PHYs respectively. Table 19 describes the clock connections to the SoC device.

Signal Name	Frequency	SoC Pin	SoC Pin Type
CLK50	50 MHz	AM5	HSIOPB8T4L6/CLKIN_N_12/CCC
MSS_REFCLK_P MSS_REFCLK_N	125 MHz	V3 W34	MSS_REFCLK_IN_P MSS_REFCLK_IN_N

Table 19: Module Clock Resources

2.12 Reset

There are three dedicated reset signals on the module; the device reset signal FPGA_DEVRST#, the peripheral reset signal RST_PER# and the MSS_RST# signal, which is the control signal of the peripheral reset.

Pulling FPGA_DEVRST# low resets the SoC device and the SPI Flash and additionally asserts RST_PER#. FPGA_DEVRST# is available on pin 132 of module connector pin A and has an on-board 4.7 kΩ pull-up resistor to VCC_CFG.

Pulling RST_PER# low resets the Ethernet PHY(s), the USB PHY, the QSPI and eMMC flash devices. RST_PER# can be asserted by pulling low MSS_RST# which is connected to the MSS of the SoC and has a 4.7 kΩ pull-up resistor to VCC_CFG.

For more details about the reset signals, consult PolarFire FPGA and PolarFire SoC FPGA Power-Up and Reset [25].

Signal Name	Connector Pin	SoC Pin	Pin Name	Description
FPGA_DEVRST#	A-132	G9	DEVRST_N	Device reset
RST_PER#	-	-	-	Peripheral reset
MSS_RST#	-	AA12	MSSIO12B4	Control signal of peripheral reset

Table 20: Reset Resources

FPGA_DEVRST# and RST_PER# are automatically asserted if PWR_GOOD is low.

2.13 LEDs

There are four active-low user LEDs on the Mercury+ MP1 SoC module - two of them are connected to the FPGA fabric only. The other two are connected to the FPGA fabric and the MSS. This results in 6 LED control signals. The LEDs will be on if at least one of its respective control signals is low.

Signal Name	SoC Pin	Bank	Bank Type	LED	Comment
FPGA_LED0#	E23	9	GPIO	LED 0	User function/active-low
MSS_LED0#	U14	2	MSS	LED 0	User function/active-low
FPGA_LED1#	D23	9	GPIO	LED 1	User function/active-low
MSS_LED1#	V12	2	MSS	LED 1	User function/active-low
FPGA_LED2#	A25	9	GPIO	LED 2	User function/active-low
FPGA_LED3#	B25	9	GPIO	LED 3	User function/active-low

Table 21: User LEDs

2.14 DDR4 SDRAM (MSS)

There are two DDR4 SDRAM channels on the Mercury+ MP1 SoC module: one attached directly to the MSS side and one attached directly to the FPGA fabric.

The DDR4 SDRAM connected to the MSS is mapped to MSS Bank 6. The memory configuration on the Mercury+ MP1 SoC module supports ECC error detection and correction; the correction code type used is single bit error correction and double bit error detection (SEC-DED).

Three 16-bit memory chips are used to build an 36-bit wide memory (12 bits are unused): 32 bits for data and 4 bits for ECC.

The maximum memory bandwidth on the Mercury+ MP1 SoC module is:
1'600 Mbit/s × 32 bit = 6'400 MB/s

2.14.1 DDR4 SDRAM Characteristics

Table 22 describes the MSS memory availability and configuration on the Mercury+ MP1 SoC module.

Module	Chip Density	Configuration
ME-MP1-D3EN	8 Gbit	512 M × 16 bit
ME-MP1-D4E	16 Gbit	1 G × 16 bit

Table 22: DDR4 SDRAM (MSS) Characteristics

2.14.2 Signal Description

Refer to the Mercury+ MP1 SoC Module FPGA Pinout Excel Sheet [4] for detailed information on the DDR4 SDRAM connections.

2.14.3 Termination

Tip

No external termination is implemented for the data signals on the Mercury+ MP1 SoC module. Enclustra strongly recommends enabling the on-die termination (ODT) feature of the assembled device.

2.14.4 Parameters

Table 23 shows the parameters of the MSS DDR4 SDRAM to be set in the Libero project such that it corresponds to the reference design [2] of the Mercury+ MP1 SoC module.

Refer to the Mercury+ MP1 SoC Module Reference Design [2] for DDR4 settings guidelines.

Parameter	Value
DDR memory type	DDR4
DQ width	32 bit
ECC	Enabled
Bank group address width	1
Row address width	16 - 17
Column address width	10
Bank address width	2
Memory clock frequency	666 MHz
Memory CAS latency	11
Memory write CAS latency	11
tRAS	35 ns
tRCD	13.75 ns
tRP	13.75 ns
tRC	48.75 ns
tWR	15 ns
tCCD_L	5 cycles
tCCD_S	4 cycles
tREFI	7.8 µs
tRFC	350 ns
tFAW	35 ns

Table 23: DDR4 SDRAM (MSS) Parameters

2.15 DDR4 SDRAM (FPGA Fabric)

The DDR4 SDRAM connected to the FPGA fabric is mapped to HSIO banks 0 and 8. The DDR bus width is 64-bit.

For standard speed grade rated SoC devices (MPFS250TS-FCG1152I) speeds up to 1’333 Mbit/s (667 MHz) are supported by the memory controller and the assembled memory devices, whereas for -1 speed grade rated SoC devices (MPFS460TS-1FCG1152I) speeds up to 1’600 Mbit/s (800 MHz) are supported by the

memory controller and the assembled memory devices.

The maximum FPGA fabric memory bandwidth on the Mercury+ MP1 SoC module using -1 speed grade rated devices is:
1'600 Mbit/s × 64 bit = 12'800 MB/s

The maximum FPGA fabric memory bandwidth on the Mercury+ MP1 SoC module using STD speed grade rated devices is:
1'333 Mbit/s × 64 bit = 10'664 MB/s

2.15.1 DDR4 SDRAM Characteristics

Table 22 describes the memory availability and configuration on the Mercury+ MP1 SoC module.

Module	Chip Density	Configuration
ME-MP1-D3EN	–	–
ME-MP1-D4E	16 Gbit	1 G × 16 bit

Table 24: DDR4 SDRAM (FPGA Fabric) Characteristics

2.15.2 Signal Description

Refer to the Mercury+ MP1 SoC Module FPGA Pinout Excel Sheet [4] for detailed information on the DDR4 SDRAM connections.

2.15.3 Termination

> **Tip**
>
> No external termination is implemented for the data signals on the Mercury+ MP1 SoC module. Enclustra strongly recommends enabling the on-die termination (ODT) feature of the assembled device.

2.15.4 Parameters

Table 25 shows the parameters of the FPGA fabric DDR4 SDRAM to be set in the Libero project such that it corresponds to the reference design [2] of the Mercury+ MP1 SoC module.

Refer to the Mercury+ MP1 SoC Module Reference Design [2] for DDR4 settings guidelines.

Parameter	Value
DDR memory type	DDR4
Memory clock frequency	666 MHz
DQ width	64 bit
Row address width	16 - 17
Column address width	10
Bank address width	2
Bank group address width	2
ECC	Disabled
Memory CAS latency	11
Memory write CAS latency	11
tRAS	35 ns
tRCD	13.75 ns
tRP	13.75 ns
tRC	48.75 ns
tWR	15 ns
tCCD_L	5 cycles
tCCD_S	4 cycles
tREFI	7.8 µs
tRFC	350 ns
tFAW	35 ns

Table 25: DDR4 SDRAM (FPGA Fabric) Parameters

2.16 QSPI Flash

The QSPI flash can be used to store application code and other user data.

2.16.1 QSPI Flash Characteristics

Table 26 describes the memory availability and configuration on the Mercury+ MP1 SoC module.

Part Number	Size	Manufacturer
S25FL512S	512 Mbit	Infineon (Spansion)

Table 26: QSPI Flash Characteristics

Tip

Different flash memory devices may be assembled in future revisions of the Mercury+ MP1 SoC module. Any flash memory with a different speed and temperature range fulfilling the requirements of the module variant may be used.

2.16.2 Signal Description

The QSPI flash is connected to MSS bank 2, pins MSSIO 30 to 35.

The reset of the QSPI flash can be asserted by either MSS_RST# or FPGA_DEVRST#.

2.16.3 Configuration

The QSPI flash supports up to 50 MHz operation for standard read. For fast, dual and quad read speed values, refer to the flash device datasheet.

Using the Write Register (WWR) command can corrupt the QSPI flash. This issue is described in more details in the Mercury+ MP1 SoC Module Known Issues and Changes [7].

2.17 SPI Flash

The SPI flash is intended for SPI master FPGA programming (Auto Update or In-Application Programming). Refer to Section 3.3 for details on programming the flash memory.

2.17.1 SPI Flash Characteristics

Table 27 describes the memory availability and configuration on the Mercury+ MP1 SoC module.

Part Number	Size	Manufacturer
S25FL512S	512 Mbit	Infineon (Spansion)

Table 27: SPI Flash Characteristics

Tip

Different flash memory devices may be assembled in future revisions of the Mercury+ MP1 SoC module. Any flash memory with a different speed and temperature range fulfilling the requirements of the module variant may be used.

2.17.2 Signal Description

The SPI flash can either be connected to the FPGA configuration pins on bank 3 or to the module connector pins via multiplexer controlled by the SRST# and SPI_FLASH_MODE signals (see Section 3.3).

The reset of the SPI flash can be asserted by FPGA_DEVRST#.

2.18 eMMC Flash

The eMMC flash can be used for booting Linux and to store application code and other user data.

2.18.1 eMMC Flash Characteristics

The Mercury+ MP1 SoC module is equipped with a 16 GB eMMC flash.

2.18.2 Signal Description

The eMMC flash signals are multiplxed with the SD card signals. If signal SDIO_SEL is set low, which is available on HSIO bank 8, the eMMC signals are routed to MSS bank 4 for 8 bit data transfer mode. The command signal has a 4.7 kΩ pull-up resistor to VCC_CFG and the data lines have 47 kΩ pull-up resistors to VCC_CFG.

2.18.3 Configuration

The following speed modes are supported, depending on VCC_CFG:

- VCC_CFG = 3.3V: Default Speed (up to 26 MHz), High Speed (up to 52 MHz)
- VCC_CFG = 2.5V: Invalid
- VCC_CFG = 1.8V: Default Speed (up to 26 MHz), High Speed (up to 52 MHz), HS200 (up to 200 MHz)

2.19 SD Card

An SDIO interface is available on module connector A. This interface allows to attach an SD card via carrier board. External pull-up resistors are needed for SD card operation. For VCC_CFG of 1.8 V, a level shifter to 3.3 V is required (some level shifters also have built-in pull-ups). A VCC_CFG of 2.5 V is not supported in combination with SD card. For booting from an Ultra High Speed (UHS) SD card, an SD 3.0 compliant level shifter is required on the base board and VCC_CFG must be set to 1.8 V. This boot mode has not been tested, but it may be supported in the future.

The SD card can be used for booting Linux and to store application code and other user data. The SD card signals are multiplexed with the eMMC signals. The signal SDIO_SEL needs to be set high to configure the multiplexer for SD card operation. This configuration routes the SDIO signals on MSS bank 4 to the module connector A.

2.20 Dual Gigabit Ethernet

Up to two 10/100/1000 Mbps Ethernet PHYs are available on the Mercury+ MP1 SoC module. Both are connected via SGMII interfaces to the Microprocessor Subsystem.

2.20.1 Ethernet PHY Characteristics

Table 28 describes the Ethernet PHY devices assembled on the Mercury+ MP1 SoC module.

Part Number	Manufacturer	Description
DP83867IS	Texas Instruments	10/100/1000 Mbps

Table 28: Gigabit Ethernet PHYs Characteristics

2.20.2 Signal Description

PHY 0 is connected to MSS Ethernet MAC 0 and PHY 1 is connected to MSS Ethernet MAC 1 through the dedicated SGMII pins at MSS bank 5.

The two Gigabit Ethernet PHYs have a shared MDIO interface and a shared interrupt line. The MDIO signals are connected to MSS bank 2, pins 28 and 29. These pins can be controlled by Ethernet MAC 1. The interrupt output of the Ethernet PHYs is connected to the I2C interrupt signal, available on MSS bank 4,

pin 13.

The product models with one Ethernet PHY only have Ethernet PHY 1 assembled.

2.20.3 External Connectivity

The Ethernet signal lines can be connected directly to the magnetics. Refer to the Mercury Mars Module Pin Connection Guidelines [11] for details regarding the connection of Ethernet signals.

2.20.4 PHY Configuration

The configuration of the Ethernet PHYs is bootstrapped when the PHYs are released from reset. The bootstrap options of the Ethernet PHYs are set as indicated in Table 29.

Strap Input	Mode	Description
RX_D0	1 4	PHY0: MDIO address 0 PHY1: MDIO address 3
RX_D2	1 4	PHY0: MDIO address 0 PHY1: MDIO address 3
RX_CTRL	3	Enable autonegotiation
GPIO_1	1	Not relevant for SGMII operation
GPIO_0	1	Not relevant for SGMII operation
LED_2	1	Not relevant for SGMII operation
LED_1	1	Advertise 10/100/1000 ability
LED_0	2	Enable SGMII / disable MDI mirroring

Table 29: Gigabit Ethernet PHYs Configuration - Bootstraps

2.21 USB 2.0

One USB 2.0 PHY is available on the Mercury+ MP1 SoC module. Is connected to the MSS to I/O bank 2 and can be configured as device or host.

2.21.1 USB PHY Characteristics

Table 30 describes the USB PHY device assembled on the Mercury+ MP1 SoC module.

Part Number	Manufacturer	Description
USB3320C	Microchip	USB 2.0 PHY

Table 30: USB 2.0 PHY Characteristics

2.21.2 Signal Description

The ULPI interface for the PHY is connected to MSS pins 14-25 for use with the integrated OTG-compliant USB core.

2.22 Real-Time Clock (RTC)

The Mercury+ MP1 SoC module features a real-time clock. The RTC can be accessed by the I2C bus.

2.22.1 RTC Characteristics

Table 31 describes the RTC device assembled on the Mercury+ MP1 SoC module.

Part Number	Manufacturer
ISL12020M	Renesas

Table 31: Real-Time Clock Characteristics

If the RTC functionality is needed, voltage VCC_BAT, which is located at pin A168 of module connector A, needs to be applied.

2.23 Secure EEPROM

The secure EEPROM is used to store the module type and serial number, as well as the Ethernet MAC address and other information. It is connected to the I2C bus.

The secure EEPROM must not be used to store user data.

Refer to Section 4.4 for details on the content of the EEPROM.

2.23.1 EEPROM Characteristics

Table 32 describes the EEPROM device assembled on the Mercury+ MP1 SoC module.

Part Number	Manufacturer
ATSHA204A-MAHDA-T (default)	Atmel
24AA128T-I/MNY (assembly option)	Microchip

Table 32: EEPROM Characteristics

An example demonstrating how to read data from the EEPROM is included in the Mercury+ MP1 SoC Module Reference Design [2].

2.24 Trusted Platform Module (TPM)

The Mercury+ MP1 SoC module is equipped with a trusted platform module (TPM). The TPM is a tamper-resistant security controller (TCG standard TPM 2.0) with advanced cryptographic algorithms (RSA-2048, ECC-256, SHA-256). For details, refer to the TPM datasheet.

The SPI interface of the TPM is connected to FPGA GPIO Bank 7.

2.24.1 TPM Characteristics

Table 33 describes the TPM assembled on the Mercury+ MP1 SoC module.

Part Number	Manufacturer
SLI9670AQ20FW1311XUMA1	Infineon

Table 33: TPM Characteristics

3 Device Configuration

3.1 Configuration Signals

Table 34 describes the most important configuration pins and their location on the module connector. These signals allow the user to program the SPI flash and SoC via SPI. Refer to Sections 3.3 and 3.7 for details.

Signal Name	SoC Pin Name	Connector Pin	Description	Comment
FPGA_DEVRST#	DEVRST_N	A-132	SoC device reset	4.7 kΩ pull-up to VCC_CFG
SRST#	HSIONB8T4L5	A-124	SPI programming mode selection (Table 35)	4.7 kΩ pull-up to VCC_CFG
SPI_FLASH_MODE	IO_CFG_INTF HSIOPB8T4L5	A-126	SPI programming mode selection (Table 35)	4.7 kΩ pull-up to VCC_CFG
SDIO_SEL	HSIONB8T4L6	-	SDIO/eMMC selection	4.7 kΩ pull-up to VCC_3V3_SEQ

Table 34: SoC Configuration Pins

3.2 Module Connector C Detection

Signal C_PRSNT# (pin C-167) must be connected to GND on the base board if the designed base board has three connectors. Depending on the value of this pin, the FPGA banks routed to module connector C are supplied with the voltages provided by the user (when C_PRSNT# is low) or with a default voltage of 3.3 V (when C_PRSNT# is unconnected).

C_PRSNT# is equipped with a 4.7 kΩ pull-up resistor on the module.

3.3 SPI Programming Modes

There are three PolarFire SoC programming modes. Two of these three modes are via the SPI interface of the SoC (referred as PF Bank 3 in Table 35), which are described in this section. The other programming mode is through the JTAG interface. For more information on the JTAG programming mode, refer to Section 3.4.

For SPI configuration modes the SoC can either act as the slave or master. On the Mercury+ MP1 SoC module a third SPI programming mode can be configured, giving the user access to the SPI flash mermory by a external master and bypassing the SoC.

Table 35 gives an overview over these three SPI programming mode configuration options which can be set via the two configuration signals SPI_FLASH_MODE and SRST#.

SPI_FLASH_MODE	SRST#	SPI Master	SPI Slave	Bitstream Source
0	0	Base board	SPI Flash	FPGA
0	1	Base board	PF Bank 3	FPGA
1	–	PF Bank 3	SPI Flash	FPGA/SPI-Flash

Table 35: SPI Programming Mode Configuration Options

For more details on Microchip PolarFire SoC programming, refer to the PolarFire® FPGA and PolarFire SoC FPGA Programming User Guide [28].

NOTICE

Damage to the device

The module can be damaged if the SPI signals are driven simultaneously by the module and the base board.

- For programming the FPGA via external SPI master, disable the "auto update" programming mode to prevent that the signals are driven simultaneously.

3.4 JTAG

PolarFire devices include a JTAG controller which is able to communicate with the PolarFire system controller and can be used for programming, system testing, debugging.

3.4.1 JTAG on Module Connector

The module connector locations of the JTAG interface are given in Table 36.

Signal Name	Connector Pin	SoC Pin	Comment
JTAG_TCK	A-123	TCK	4.7 kΩ pull-down to GND
JTAG_TMS	A-119	TMS	4.7 kΩ pull-up to VCC_CFG
JTAG_TDI	A-117	TDI	4.7 kΩ pull-up to VCC_CFG
JTAG_TDO	A-121	TDO	-

Table 36: JTAG Interface

3.4.2 External Connectivity

The JTAG signals can be connected to a JTAG connector on the base board. No pull-up/pull-down resistors are necessary. The VJTAG pin of the programmer must be connected to VCC_CFG.

Enclustra recommends to add series termination resistors between the module and the JTAG connector, close to the source. Enclustra also recommends adding a transient-voltage-suppression diode close to the JTAG connector to protect the MPSoC from damage due to ESD (Electrostatic Discharge). Refer to the Mercury Mars Module Pin Connection Guidelines [11] for details on JTAG interface.

Tip

On Enclustra base boards, the pin assignment of the JTAG connector is different than the Microchip FlashPro Programmer. The appropriate connection is described in the Mercury+ MP1 SoC Module Reference Design [2].

3.4.3 JTAG Programming Mode

The PolarFire SoC can be programmed via the JTAG interface. In JTAG programming mode, the SoC can be programmed by the use of a FlashPro Programmer or an external Microprocessor. For details, refer to the PolarFire® FPGA and PolarFire SoC FPGA Programming User Guide [28].

3.5 eMMC Flash Programming

The USB service in Hart Software Services (HSS) software can be used to program the eMMC memory. With the USB PHY assembled on the module configured in device mode, the eMMC (or SD card) can be used as USB flash drive. This service allows to format and partition the eMMC by any host computer.

See Microchip documentation or the Linux reference design documentation [17] for more details.

3.6 SPI Flash Programming via JTAG

The Microchip Libero SoC Design Suite software offers SPI flash programming support via JTAG. To program the SPI flash using JTAG, the multiplexer must be configured as described in Section 3.3.

3.7 SPI Flash Programming from an External SPI Master

To program the SPI flash from an external SPI master, the multiplexer must be configured as described in Section 3.3. Table 37 describes the QSPI flash signals used for external access.

Signal Name	Connector Pin	Description	Comment
FLASH_CLK	A-118	SoC SCK	4.7 kΩ pull-up to VCC_CFG
FLASH_DO	A-122	SoC SDO	–
FLASH_DI	A-114	SoC SDI	4.7 kΩ pull-up to VCC_CFG
FLASH_CS#	A-116	SoC SS	4.7 kΩ pull-up to VCC_CFG

Table 37: QSPI Flash Signals for External Access

3.8 Enclustra Module Configuration Tool

In combination with an Enclustra base board, the QSPI flash can be programmed using Enclustra Module Configuration Tool (MCT) [19]. The Mercury+ MP1 SoC moduleis supported starting with version 2.10. The entire procedure is described in the reference design documentation.

The internal flash of the Mercury+ MP1 SoC moduleis available to be programmed trought the SPI bus, but currently cannot be programmed with the MCT. The MCT, however, can be used to erase the contents of the internal flash starting with MCT version 2.10. Due to limitations of the ST1 baseboard, the SPI communication from MCT to SoC is one-way only and procedures using the SPI bus are not available.

4 I2C Communication

4.1 Overview

The I2C bus on the Mercury+ MP1 SoC module is connected to the SoC device, the EEPROM, to the DC/DC power converters, to the RTC and is available on the module connector. This allows external devices to read the module type and to connect more devices to the I2C bus.

The I2C clock frequency should not exceed 400 kHz.

> **Tip**
>
> Maximum I2C speed may be limited by the routing path and additional loads on the base board.

> **Tip**
>
> If the I2C traces on the base board are very long, 100 Ω series resistors should be added between the module and the I2C device on the base board.

4.2 Signal Description

Table 38 describes the signals of the I2C interface - the pins are connected to both MSS and FPGA fabric. All signals have on-board pull-up resistors to VCC_3V3.

All signals must be connected to open collector outputs and must not be driven high from any source. I2C_INT# is an open drain signal, has a pull-up resistor to VCC_CFG and must not be driven high by the SoC.

Level shifters are used between the I2C bus and SoC pins, as MSS bank 2, MSS bank 4 and FPGA GPIO bank 1 are supplied with different voltages. Ensure that all pins are configured correctly and no pull-down resistors are enabled.

Signal Name	MSS Pin	SoC Pin	Connector Pin	Comment
I2C_SDA	MSSIO27	V13	A-113	2.2 kΩ pull-up
I2C_SCL	MSSIO26	V14	A-111	2.2 kΩ pull-up
I2C_INT#	MSSIO13	Y12	A-115	4.7 kΩ pull-up

Table 38: I2C Signal Description

4.3 I2C Address Map

Table 39 describes the addresses for several devices connected on I2C bus.

Address (7-bit)	Description
0x64	Secure EEPROM
0x54	Secure EEPROM (assembly option, refer to Section 2.23)
0x57	RTC user SRAM
0x6F	RTC registers
0x11	Power Converter
0x17	Power Converter

Table 39: I2C Addresses

4.4 Secure EEPROM

The secure EEPROM is used to store the module serial number and configuration. The functionality to read data from the EEPROM is included in the Enclustra Yocto BSP Layer [17].

> **Tip**
>
> Any attempt to write data to the secure EEPROM causes the warranty to be rendered void.

4.4.1 Memory Map

Address	Length (bits)	Description
0x00	32	Module serial number
0x04	32	Module product information
0x08	40	Module configuration
0x0D	24	Reserved
0x10	48	Ethernet MAC address
0x16	48	Reserved
0x1C	32	Checksum (only for DS28CN01 EEPROM type)

Table 40: EEPROM Sector 0 Memory Map

Module Serial Number

The module serial number is a unique 32-bit number that identifies the module. It is stored using big-endian byte order (MSB on the lowest address).

Module Product Information

This field indicates the type of module and hardware revision.

Module	Product Series	Reserved	Revision	Product Information
Mercury+ MP1 SoC module	0x0339	0x[XX]	0x[YY]	0x0339 [XX][YY]

Table 41: Product Information

Module Configuration

Addr.	Bits	Description	Min. Value	Max. Value	Comment
0x08	[7:4]	SoC type	0	2	See SoC type table (Table 43)
	[3:0]	SoC device speed grade	0	1	See SoC type table (Table 44)
0x09	[7:6]	Temperature range	0	2	See temperature range table (Table 45)
	[5]	Reserved	-	-	-
	[4:3]	Gigabit Ethernet port count	0	2	-
	[2:0]	Reserved	-	-	-
0x0A	[7:4]	eMMC flash size (GB)	0 (0 GB)	5 (16 GB)	Resolution = 1
	[3:2]	Reserved	-	-	-
	[1:0]	USB 2.0 port count	0	1	-
0x0B	[7:4]	DDR4 ECC RAM (MSS) size (GB)	0 (0 GB)	3 (4 GB)	Resolution = 1 GB
	[3:0]	DDR4 RAM (FPGA fabric) size (GB)	0 (0 GB)	4 (8 GB)	Resolution = 1 GB
0x0C	[7:4]	QSPI MSS flash size (MB)	0 (0 MB)	7 (64 MB)	Resolution = 1 MB
	[3:0]	SPI FPGA flash size (MB)	0 (0 MB)	7 (64 MB)	Resolution = 1 MB

Table 42: Module Configuration

The memory sizes are defined as Resolution$\times 2^{(\text{Value-1})}$, for example:

- DRAM = 0: none
- DRAM = 1: 1 GB
- DRAM = 2: 2 GB
- DRAM = 3: 4 GB

Table 43 shows the available SoC types.

Value	SoC Device Type
0	MPFS250T ES
1	MPFS250TS
2	MPFS460TS

Table 43: SoC Device Types

Table 44 shows the available speed grade.

Value	Module Speed Grade
0	-1
1	Standard

Table 44: Module Speed Grade

Table 45 shows the available temperature ranges.

Value	Module Temperature Range
0	Industrial
1	Extended
2	Engineering Sample

Table 45: Module Temperature Range

Ethernet MAC Address

The Ethernet MAC address is stored using big-endian byte order (MSB on the lowest address). Each module is assigned two sequential MAC addresses; only the lower one is stored in the EEPROM.

5 Operating Conditions

5.1 Absolute Maximum Ratings

Table 46 indicates the absolute maximum ratings for the Mercury+ MP1 SoC module. The values given are for reference only. For details, refer to the PolarFire SoC Datasheet [29].

Parameter	Description	Min.	Max.	Unit
VCC_MOD	Supply voltage relative to GND	-0.3	16.0	V
VCC_BAT	Supply voltage for battery-backed RAM and battery-backed RTC	-0.3	6.0	V
VCC_IO_B1 VCC_IO_B9	Output drivers supply voltage relative to GND Output drivers supply voltage relative to GND	-0.5	3.6	V
VCC_IO_B7	Output drivers supply voltage relative to GND	-0.3	3.6	V
VCC_CFG	MSSIO supply voltage relative to GND	-0.5	3.6	V
GPIO_B[x]_<...>	GPIO signal lines input voltage relative to GND	-0.5	3.8	V
IO_B[x]_<...>	HSIO signal lines input voltage relative to GND.	-0.5	3.6	V
MSSIO[x]_<...>	MSS I/O signal lines input voltage relative to GND	-0.5	3.6	V
XCVR_B[x]	Transceiver signal lines input voltage relative to GND	-0.5	1.26	V
XCVR_B[x]_REFCLK[x]	Transceivers' reference clock signals relative to GND	-0.5	3.6	V
Temperature[5]	Temperature range for extended temperature modules (E) Temperature range for industrial modules (I)	0 -40	+85 +85	°C °C

Table 46: Absolute Maximum Ratings

NOTICE

Damage to the device due to overheating

Depending on the user application, the Mercury+ MP1 SoC module may consume more power than can be dissipated without additional cooling measures.

- Ensure that the SoC is always adequately cooled by installing a heat sink and/or providing air flow.

[5] The components used on the hardware are specified for the relevant temperature range. The user must provide adequate cooling in order to keep the temperature of the components within the specified range.

5.2 Recommended Operating Conditions

Table 47 indicates the recommended operating conditions for the Mercury+ MP1 SoC module. The values given are for reference only. For details, refer to the PolarFire SoC Datasheet [29].

Parameter	Description	Min.	Max.	Unit
VCC_MOD	Supply voltage relative to GND	5.0	13.2	V
VCC_BAT	Supply voltage for battery-backed RAM and battery-backed RTC	1.8	5.5	V
VCC_IO_B[x] VCC_CFG	Output drivers supply voltage relative to GND MSSIO supply voltage relative to GND	Refer to Section 2.8.5		
GPIO_B[x]_<...>	GPIO signal lines input voltage relative to GND	0.0	VCC_IO_B[x]	V
IO_B[x]_<...> MSSIO[x]_<...>	I/O input voltage relative to GND MSS I/O signal lines input voltage relative to GND	0.0	VCC_CFG	V
Temperature[6]	Temperature range for extended temperature modules (E) Temperature range for industrial modules (I)	0 -40	+85 +85	°C °C

Table 47: Recommended Operating Conditions

NOTICE

Damage to the device due to overheating

Depending on the user application, the Mercury+ MP1 SoC module may consume more power than can be dissipated without additional cooling measures.

- Ensure that the SoC is always adequately cooled by installing a heat sink and/or providing air flow.

[6] The components used on the hardware are specified for the relevant temperature range. The user must provide adequate cooling in order to keep the temperature of the components within the specified range.

[7] For revisions prior to R3.0, the minimum value is 2.7 V.

6 Ordering and Support

6.1 Ordering

Use the Enclustra online request/order form for ordering or requesting information:
http://www.enclustra.com/en/order/

6.2 Support

Follow the instructions on the Enclustra online support site:
http://www.enclustra.com/en/support/

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